WHAT IS DATA ENGINEERING?

Concept

- Raw data is valuable but not usable in its native state.
- Data is messy, voluminous, and needs to be processed to become a valuable product.
- Data Engineers build the structure that makes this possible.
- Without this infrastructure, data science and analytics cannot happen at scale.

DATA LIFECYCLE

Data Lifecycle – Overview

- Data follows a predictable path from its creation to its final use.
- This path provides a crucial framework for all data engineering work.

GENERATION

- Data are a collection of discrete or continuous values that convey information, describing the quantity, quality, fact, statistics, other basic units of meaning, or simply sequences of symbols that may be further interpreted formally.

RETRIEVAL

Gathering data can be accomplished through:

- **Primary source**: the researcher is the first person to obtain the data.
- **Secondary source**: the researcher obtains data that has already been collected by
  other sources, such as data disseminated in a scientific journal.

Data source	Owned by	Accuracy	Use case	Privacy risk
First-party	The business	High	Personalization, retargeting	Low
Second-party	Partner	Moderate	Partnership campaigns	Moderate
Third-party	External entity	Low	Broad targeting	High

INGESTION

The process of moving data from its source into a centralized storage system.

Batch Ingestion

- Higher latency (minutes to hours)
- More efficient for large volumes
- Easiest to implement and debug
- Lower infrastructure costs

Streaming Ingestion

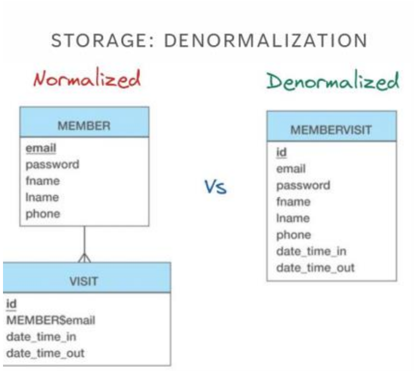
- Low latency (seconds to milliseconds)
- More complex to implement
- Higher infrastructure costs
- Enables real-time analytics

Micro-Batch Ingestion

- Near real-time processing (typically 5–15 minutes)
- Balances latency and efficiency
- Easier than true streaming
- Good for most use cases

STORAGE

- Persistently storing the ingested data for future use.
- Structured or Unstructured
- Normalized or Denormalized

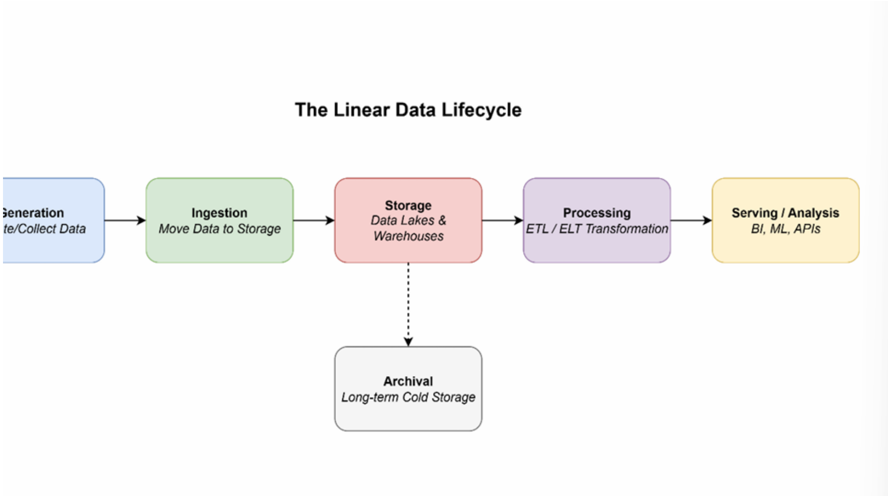


PROCESSING

Clean, Validate, Aggregate, Enrich, Transform

SERVING

Data Warehouse, Data Lakehouse, Data Mesh, API, Flat files by email?, Dashboards?



EL, ETL OR ELT

SUBJECT CURRICULUM

E – Extract

- Reading and pulling data from one or more source systems.
- Sources: Databases (SQL, NoSQL), APIs, SaaS applications (e.g., Salesforce), Log files

T – Transform

- The "business logic" step. Cleaning, enriching, and restructuring data.

Examples:

- Joining tables, Aggregating values, Masking PII, Converting date formats

L – Load

- Writing the processed data into a target destination.
- Targets:
  - Data Warehouse, Data Lake, Data Mart

EL (Extract–Load)

Examples:

- **Database Replication**: Creating a read-replica of a production database for analytical queries
  to avoid impacting production performance.
- **Data Lake Ingestion**: Moving raw files (logs, images, JSON) into a data lake like S3 or ADLS
  for long-term storage or future processing.
- **Backup and Archiving**: Simply moving data from one location to another for safekeeping.

Classic ETL

1. Extract data from sources.
2. Move it to a separate staging area or processing server.
3. Transform the data (clean, join, aggregate). This is often a resource-intensive step.
4. Load the clean, structured data into the target Data Warehouse.

ETL: PROS & CONS

Pros

- Mature & Well-Established: Tools are robust.
- Compliance-Friendly: PII can be scrubbed before loading.
- Performs Complex Logic: Optimized for heavy transformations.
- Reduces Load on Warehouse: Warehouse only handles queries.

Cons

- Slow & Rigid: Adding a new field can take weeks.
- Brittle: Transformations can break with source changes.
- High Maintenance: Requires specialized skills to manage.
- Not Ideal for Raw Data: Raw data is discarded after transformation.

(Modern) ELT

1. Extract data from sources.
2. Load the raw, untouched data directly into the target system (usually a cloud data warehouse
   like Snowflake, BigQuery, or Redshift).

3. Transform the data inside the warehouse using its powerful processing engine (e.g., using SQL).

ELT: PROS & CONS

Pros

- Speed & Flexibility: Load raw data in minutes, not hours.
- Leverages Cloud Power: Uses the scalable compute of the warehouse.
- "Schema-on-Read": No need to define structure upfront.
- All Data is Available: Analysts can access raw data for new insights.

Cons

- Potential for a "Data Swamp": Requires good governance.
- Cost Management is Key: Pay-per-query models can get expensive.
- PII is Loaded: Sensitive data enters the warehouse raw.
- Transformation logic is in SQL: Can become complex to manage.

ETL vs ELT: Comparison

Feature	ETL (Extract, Transform, Load)	ELT (Extract, Load, Transform)
Transform Location	Staging Server / ETL Tool	Target Data Warehouse
Data Structure	Schema-on-Write (Structure first)	Schema-on-Read (Flexibility first)
Load Speed	Slower (waits for transform)	Faster (loads raw data immediately)
Data Availability	Only transformed data is available	Raw + Transformed data are available
Infrastructure	On-premise or Cloud	Primarily Cloud-based
Ideal Use Case	Structured data, strict compliance, complex pre-defined reports	Big data, semi-structured sources, exploratory analysis, speed

Feature	ETL (Extract, Transform, Load)	ELT (Extract, Load, Transform)
Example Tools	Informatica, Talend, SSIS	Fivetran, dbt, Snowflake, BigQuery

DATA GOVERNANCE

WHAT IS DATA GOVERNANCE

- Formal management of data assets throughout their lifecycle
- Set of processes, policies, and standards ensuring data quality, security, and compliance

Key Objectives:

- Data Quality & Integrity
- Data Security & Privacy
- Regulatory Compliance
- Data Accessibility & Usability

WHY SHOULD WE CARE

Business Impact:

- Poor data costs organizations $15M annually (average)
- 40% of business initiatives fail due to data quality issues

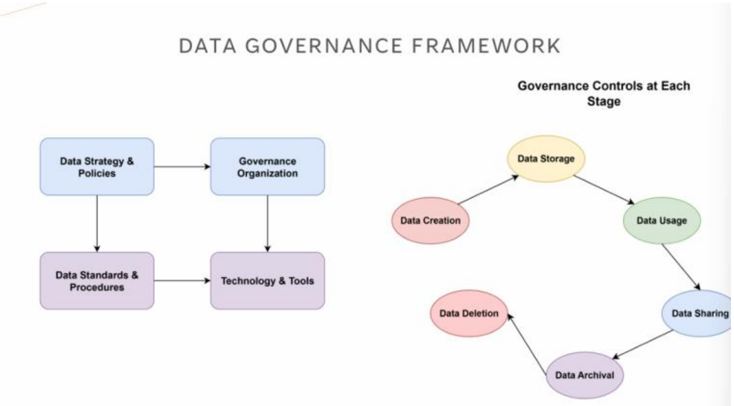
Engineering Challenges:

- Data pipeline failures due to schema changes
- Inconsistent data formats across systems
- Security vulnerabilities in data flows
- Compliance violations (GDPR, CCPA)

DATA GOVERNANCE CORE PILLARS

- **Data Quality Management**

    - Accuracy, Completeness, Consistency, Timeliness
- **Data Security & Privacy**
    - Access controls, Encryption, PII protection
- **Data Lifecycle Management**
    - Creation, Storage, Usage, Archival, Deletion
- **Metadata Management**
    - Data lineage, Cataloging, Documentation



KEY ROLES

- **Data Governance Council**
    - Strategic oversight and policy decisions. Este está por encima de los otros,
- **Data Owners**
    - Business accountability for specific datasets
- **Data Stewards**
    - Day-to-day data management and quality
- **Data Engineers**
    - Implement governance controls in pipelines
    - Ensure data lineage tracking
    - Maintain data quality checks

GOVERNANCE TASKS FOR DATA ENGINEERING

- **Pipeline Development:**
    - Schema validation and evolution
    - Data quality checks at ingestion
    - Error handling and monitoring
- **Infrastructure:**
    - Access control implementation
    - Audit logging
    - Data encryption in transit/rest
- **Documentation:**
    - Data lineage mapping
    - Pipeline documentation
    - Incident response procedures

CHALLENGES

Technical Challenges:

- Legacy system integration
- Real-time governance at scale
- Cross-platform data lineage

Organizational Challenges:

- Resistance to change
- Lack of data literacy
- Siloed data ownership

Solutions:

- Start small with pilot projects
- Automate governance controls
- Provide training and support

BEST PRACTICES

- **Build Governance into Pipelines**

    - Automated data quality tests
    - Schema validation
    - Lineage tracking
- **Implement Data Contracts**
    - Clear SLAs between teams
    - Versioned schemas
    - Breaking change notifications
- **Monitor & Alert**
    - Data quality metrics
    - Pipeline health dashboards
    - Anomaly detection

GDPR

The General Data Protection Regulation (GDPR) is a comprehensive data protection law
that came into effect on May 25, 2018. It applies to:

- All organizations processing personal data of EU residents.
- Any organization with operations in the EU.
- Any organization offering goods/services to EU residents:
  Even non-EU companies must comply if they handle EU citizens' data.

GDPR: KEY DEFINITIONS

Personal Data

Any information relating to an identified or identifiable natural person, including:

- Direct identifiers: Name, ID numbers, email addresses
- Indirect identifiers: IP addresses, location data, online identifiers
- Special categories: Health, biometric, genetic, political opinions, religious beliefs
- Pseudonymized data: Still considered personal data under GDPR

Data Controller vs. Data Processor

- **Controller**: Determines purposes and means of processing (your organization)

- **Processor**: Processes data on behalf of controller (cloud providers, vendors)

GDPR: PRINCIPLES

Technical Measures

- ☐ Encryption at rest and in transit
- ☐ Pseudonymization where possible
- ☐ Access controls and authentication
- ☐ Data backup and recovery
- ☐ Automated data deletion capabilities

Organizational Measures

- ☐ Privacy policies and procedures
- ☐ Staff training on GDPR
- ☐ Data Protection Officer (if required)
- ☐ Vendor agreements (Data Processing Agreements)
- ☐ Incident response procedures

Red Flags to Avoid

- Processing without legal basis
- Keeping data longer than necessary
- Insufficient security measures
- No process for data subject requests
- Sharing data without proper agreements
- No documentation of processing activities
- Ignoring breach notification requirements
- Invalid consent mechanisms

Data Subject Request Procedures

- Implement systems to verify identity
- Create processes to search across all systems
- Establish workflows for each type of request
- Set up tracking and response mechanisms

GDPR: WRAP UP

1. GDPR applies to most organizations handling EU residents' data
2. Personal data definition is broad – includes identifiers and pseudonymized data
3. Must have legal basis for all personal data processing
4. Individual rights are extensive – prepare systems to respond
5. Security and privacy by design are mandatory
6. Documentation is crucial for demonstrating compliance
7. Penalties are significant – up to 4% of global revenue
8. Ongoing compliance – not a one-time project

Remember: GDPR is about protecting individuals' fundamental rights while enabling legitimate business operations.
Focus on building trust through transparent, secure, and respectful data practices.